

In SettingsFragment.

```
try {
    Class<?> cls = settingsFragment.getClass();
    Method declaredMethod = HomeFragment.class.getDeclaredMethod("a1", String.class);
    Method declaredMethod2 = cls.getDeclaredMethod("a2", String.class);
    declaredMethod.setAccessible(true);
    declaredMethod2.setAccessible(true);
    HomeFragment homeFragment = new HomeFragment();
    Object objInvoke = declaredMethod2.invoke(settingsFragment, SettingsFragmentKt.getJ2());
    Intrinsics.checkNotNull(objInvoke, "null cannot be cast to non-null type kotlin.String");
    String str = SettingsFragmentKt.getJ5() + "6320" + SettingsFragmentKt.getS();
    Object objInvoke2 = declaredMethod.invoke(homeFragment, (String) objInvoke);
    Intrinsics.checkNotNull(objInvoke2, "null cannot be cast to non-null type kotlin.String");
    String str2 = (String) objInvoke2;
    if (!Intrinsics.areEqual(declaredMethod.invoke(homeFragment, string2), str) && Intrinsics.areEqual(declaredMethod.invoke(homeFragment, string2),
SettingsFragmentKt.getJ8()) && !Intrinsics.areEqual(str2, "")) {
        Context contextRequireContext = settingsFragment.requireContext();
        Object objInvoke3 = declaredMethod2.invoke(settingsFragment, "1c16" + SettingsFragmentKt.getJ5());
        Intrinsics.checkNotNull(objInvoke3, "null cannot be cast to non-null type kotlin.String");
        Toast.makeText(contextRequireContext, (String) objInvoke3, 1).show();
    }
    return;
}
```

Here `string2` is our input. The logic is basically if `a2(string2) != str` and `a2(string2) == getJ8()`, then output `a2(1c16+getJ5())`.

We can know what `getJ8()`, `getS()` and so on are in `SettingsFragmentKt`.

```
3 public final class SettingsFragmentKt {
9     private static final String f = "43";
0     private static final String j2;
1     private static final String j5;
2     private static final String j8;
3     private static final String s = "69";
4     private static final String u;
5     private static final String j1 = "1c16051b09213e33292e283b392e333534053c2f34342327";
6     private static final String j9 = "1334393528283f392e7a293f3928";
7
8     public static final String getS() {
9         return s;
0     }
1
2     public static final String getF() {
3         return f;
4     }
5
6     static {
7         String str = "2f3436353931";
8         u = str;
9         j8 = str + "05293f39283f2e05626e696b";
0         j5 = "1b1d21293f39283f2e05" + str + "3f3e05626e696b27";
1         j2 = "1c16051" + str + "83b392e333534053c2f34342327";
2     }
3
4     public static final String getU() {
5         return u;
6     }
7
8     public static final String getJ8() {
9         return j8;
0     }
1
2     public static final String getJ5() {
3         return j5;
4     }
5
6     public static final String getJ1() {
7         return j1;
8     }
9
0     public static final String getJ2() {
1         return j2;
2     }
3 }
```

So j5 is 1b1d21293f39283f2e052f34363539313f3e05626e696b27.

a2 firstly cuts string into 2 bytes a group. Then it parses them as hex into integer.

```
private final String a2(String hex) {
    try {
        List<String> listChunked = StringsKt.chunked(hex, 2);
        ArrayList<String> arrayList = new ArrayList();
        Iterator<String> it = listChunked.iterator();
        while (it.hasNext()) {
            Object objInvoke = Integer.class.getMethod("parseInt", String.class, Integer.TYPE).invoke(null, it.next(), 16);
            Intrinsic.checkNotNull(objInvoke, "null cannot be cast to non-null type kotlin.Int");
            int iIntValue = ((Integer) objInvoke).intValue();
            Method declaredMethod = getClass().getDeclaredMethod("xorOp", Integer.TYPE, Integer.TYPE);
            declaredMethod.setAccessible(true);
            Object objInvoke2 = declaredMethod.invoke(this, Integer.valueOf(iIntValue), 90);
            Intrinsic.checkNotNull(objInvoke2, "null cannot be cast to non-null type kotlin.Int");
            arrayList.add(Byte.valueOf((byte) ((Integer) objInvoke2).intValue()));
        }
        Object objNewInstance = String.class.getConstructor(byte[].class, Charset.class).newInstance(CollectionsKt.toByteArray(arrayList), Charsets.UTF_8);
        Intrinsic.checkNotNull(objNewInstance, "null cannot be cast to non-null type kotlin.String");
        return (String) objNewInstance;
    } catch (Exception unused) {
        return "";
    }
}
```

Every integer xor 90, then added into a list. 90 is also 0x5a.

```
int iIntValue = ((Integer) objInvoke).intValue();
Method declaredMethod = getClass().getDeclaredMethod("xorOp", Integer.TYPE, Integer.TYPE);
declaredMethod.setAccessible(true);
Object objInvoke2 = declaredMethod.invoke(this, Integer.valueOf(iIntValue), 90);
Intrinsic.checkNotNull(objInvoke2, "null cannot be cast to non-null type kotlin.Int");
arrayList.add(Byte.valueOf((byte) ((Integer) objInvoke2).intValue()));
```

So now we can calculate a2(1c16+getJ5()).

I wrote a script for this:

```
1 u = "2f3436353931"
2 j5 = "1b1d21293f39283f2e05" + u + "3f3e05626e696b27"
3
4 data = "1c16" + j5
5 chunk = []
6 for i in range(0, len(data), 2):
7     tmp = data[i:i+2]
8     chunk.append(int(tmp, 16))
9
10 flag = ""
11 for v in chunk:
12     flag += chr(v ^ 0x5a)
13 print(flag)
```

The result is:

```
(mahiro@kali) - [~/Homework/defense/settings/a2]  
● $ python3 exploit.py  
FLAG{secret_unlocked_8431}
```

FLAG{secret_unlocked_8431}